

ContextLens

An Evidence-Grounded Agent for Verifiable Historical Address Investigation

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Duke Kunshan University (DKU)

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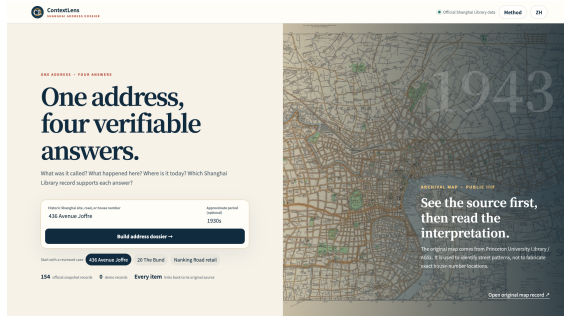
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Why historical addresses are difficult to verify

Dated place names change meaning over time

- Street names change across time.
- Numbers and landmarks remain ambiguous.
- Archival coverage is uneven.
- Fluent answers exceed available evidence.

Research Question: How can an agent resolve a dated place before retrieval and keep each claim inspectable?



The interface starts from one address and asks four source-backed questions.

Place identity resolution before retrieval

Core design

A dated address becomes a typed, replayable investigation.

01 Temporal place resolution

Dated aliases and competing candidates remain visible.

02 Relation and claim gates

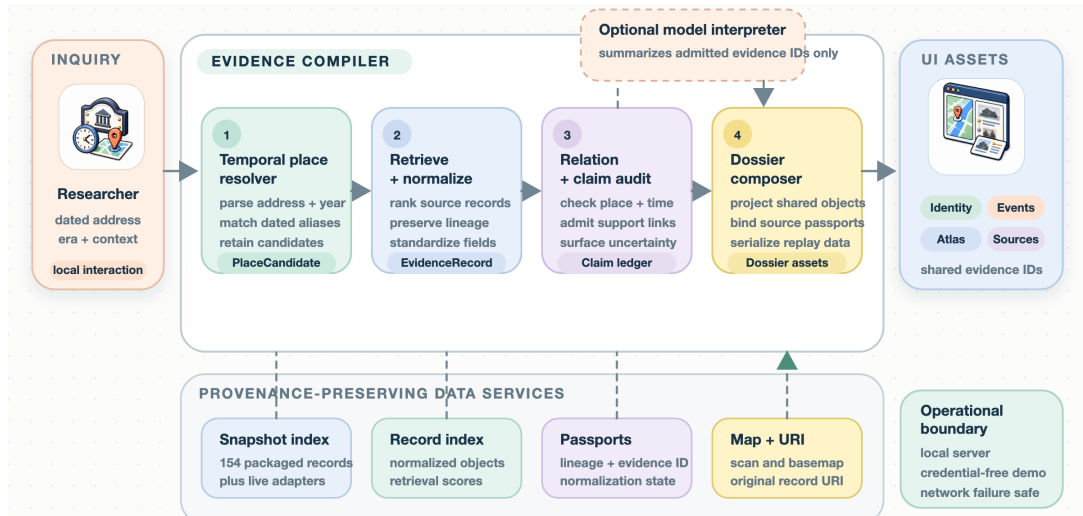
Only place- and time-compatible evidence can support a claim.

03 Source Passports

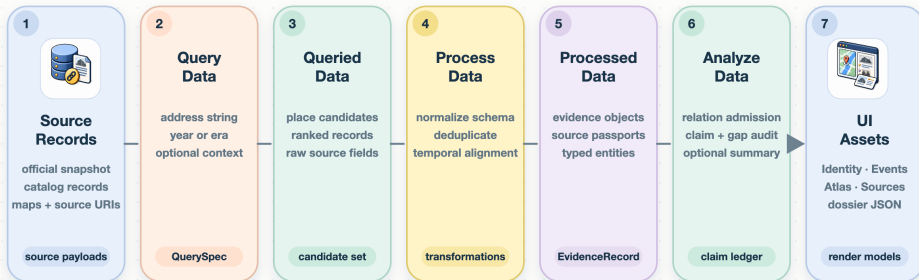
Provider, dataset, period, evidence ID, normalization state, and original URI travel with each record.



Evidence-grounded architecture



Provenance across the data pipeline



Provenance spine

Identifiers and source boundaries persist through every stage.

— evidence_id · source_uri · temporal scope · normalization state →

Evidence ID, source URI, temporal scope, and normalization state persist across every stage.

Demo case: 436 Avenue Joffre in 1934

[← New search](#)

Avenue Joffre (now Huaihai Middle Road)

Built 3 spatiotemporal nodes and 2 verifiable claims around Avenue Joffre (now Huaihai Middle Road), covering 1927–1934.

[Print / save PDF](#)[Download evidence](#)

01 **Address identity**
Old name → current name

02 **What happened**
In chronological order

03 **Then and now**
Original map + modern map

04 **Evidence sources**
Open every record

ANSWER 01 • ADDRESS IDENTITY

Avenue Joffre



Huaihai Middle Road

Exact match on the historic road name; historical event records explicitly identify Avenue Joffre as today's Huaihai Middle Road.

Address resolution confidence **96%**

How this road acquired its current name

01 **Xijiang Road**
1901—1906

02 **Baochang Road**
1906—1915

03 **Avenue Joffre**
1915—1943

04 **Taishan Road**
1943—1945

05 **Linsen Middle Road**
1945—1950

06 **Huaihai Middle Road**
1950—present

WHAT WE CAN SAY

2 direct claims

Founding of Kangjian Bookstore has a verifiable place–event relationship with Avenue Joffre (1934–1950).
[View evidence](#)

At least 3 independent sources connect Avenue Joffre (now Huaihai Middle Road) to events or buildings from different periods.
[View evidence](#)

Nothing unsupported by a source is filled in as a story.

Current code capture, 10 September 2026: 96% address-resolution confidence, six dated names, and two direct claims.

6

Evidence-bounded events and spatial context

Three evidence-linked nodes

The screenshot shows the ContextLens interface for 'VERIFIED ADDRESS DOSSIER'. The search bar contains 'Avenue Joffre (now Huaihai Middle Road)'. Below the search bar, there are three tabs: 'Address Identity', 'What happened', and 'Then and now'. The 'What happened' tab is selected, showing a list of events. The first event is '1927 Liu Haisu Recent Works Exhibition' at 'Shangshan Hall, Avenue Joffre'. The second event is '1934-1950 Founding of Kangjian Bookstore' at '436 Avenue Joffre'. The third event is 'Undated Cathay Theatre' under the 'Date under review' section.

ContextLens
VERIFIED ADDRESS DOSSIER

← New search Avenue Joffre (now Huaihai Middle Road)
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01 Address Identity
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Original map → modern map

ANSWER 02 - WHAT HAPPENED HERE

3 verifiable place nodes

Ordered by dates in the records; items marked "date under review" are not forced into a definite timeline.

1927 Event
Liu Haisu Recent Works Exhibition
Shangshan Hall, Avenue Joffre
View original evidence →

1934-1950 Event
Founding of Kangjian Bookstore
436 Avenue Joffre
View original evidence →

Date under review Building
Cathay Theatre

1927: Liu Haisu exhibition

1934-1950: Kangjian Bookstore

Undated: Cathay Theatre under review

Two distinct map roles

The screenshot shows the ContextLens interface for 'VERIFIED ADDRESS DOSSIER'. The search bar contains 'Avenue Joffre (now Huaihai Middle Road)'. Below the search bar, there are four tabs: 'Address Identity', 'What happened', 'Then and now', and 'Evidence sources'. The 'Then and now' tab is selected, showing two maps side-by-side. The left map is a historical map from 1943, and the right map is a present-day map from 2026. The text 'Read the historical original beside today's location.' is displayed above the maps. Below the maps, there is a note: 'An unfinished georeferencing is not presented as a precise overlay. The source map is on the left; today's location is on the right.'

ContextLens
VERIFIED ADDRESS DOSSIER

← New search Avenue Joffre (now Huaihai Middle Road)
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Original map → modern map

04 Evidence sources
Open every record

ANSWER 03 - THEN AND NOW

Read the historical original beside today's location.

An unfinished georeferencing is not presented as a precise overlay. The source map is on the left; today's location is on the right.

1943 Historical original

2026 Present-day location

The 1943 scan supports visual orientation. The current map locates the resolved address. Exact house-number alignment remains unverified.

Source Passports link claims to original records

A claim and its evidence card share one identifier

For the Kangjian Bookstore claim, the interface exposes:

- Shanghai Library as the provider.
- the Historical and Cultural Events dataset.
- the 1934–1950 temporal scope.
- evidence ID event–kangjian–1934.
- normalization state and original URI.

The normalized record preserves the source boundary.



Explicit ambiguity and unresolved states

The system treats uncertainty as an output state

Input state	Example	System response	Claim behavior
Resolved	436 Avenue Joffre, 1934	One temporally compatible candidate	Admit two direct claims
Ambiguous	Nanjing Road depart- ment store	Retain East and West Nanjing Road candidates	Request more context
Unresolved	9999 Mars Road	Return unresolved and ask for evidence	Generate no historical fact

No candidate, no claim. Competing candidates remain available for review.

Implementation and verification

154

verified snapshot records

0

demo-seed records on the
reviewed path

29

local Python test functions passed

Implementation

- Python controller with a deterministic evidence compiler
- TypeScript UI, local HTTP API, and SQLite evidence index
- Credential-free local run with optional live adapters
- Model assistance occurs after relation and claim auditing

Verified behavior

- dated address resolution, ambiguity, and negative cases
- relation gating and evidence-bounded model citations
- network failure, TLS, and privacy-safe audit logs
- asynchronous states and interface contracts

Interpretation: These checks verify implementation behavior. They do not establish retrieval accuracy or user-study outcomes.

Supported in the current demo

- Shanghai-specific packaged snapshot
- inspectable evidence and Source Passports
- explicit ambiguity and unresolved states
- offline interaction and failure-safe dossier output

Open research questions

- resolution quality across a broader address benchmark
- expert correction workflows and historian evaluation
- claim-citation precision and contradiction handling
- more precise georeferencing for historical maps

ContextLens is an investigation interface, not a general historical geocoder.

Some coordinates remain coarse. The system has not yet completed a historian or public-user study.

Scholarship support

Sean Wan, Shilin Ou, Yutian Wang, and Polina Postnikova acknowledge support from the Summer Research Scholars Program at Duke Kunshan University (DKU) and from the DKU Library grant supporting the Shanghai Library Open Data Contest project entitled **StableTradeAtlas**, both under the supervision of Prof. Luyao Zhang.

Appendix

StableTradeAtlas Team Profiles



DKU Team Profile

Project Lead / Agent Architecture / Trustworthy Evaluation



Sean Wan

Project leadership and
trustworthy agent
evaluation

Relevant expertise

- Agentic AI evaluation and benchmarks
- RAG workflows and structured answers
- Cost, latency, citation, and risk auditing
- Full-stack MVP and GitHub coordination

Contribution highlight

Sean leads StableTrade Atlas's trustworthy-agent architecture. His StableEval Arena experience informs prediction-quality, risk, and evidence-aware evaluation.

Key contributions

- **Historical Evidence Compiler:**
Connects old Shanghai addresses with traceable people, places, events, and documents.
- **Trustworthy AI:** Adds claim-level citations, uncertainty auditing, privacy protection, and evidence-based guardrails.
- **Spatiotemporal public experience:**
Integrates historical maps, modern 3D geography, and bilingual interactive storytelling.



DKU Team Profile: Shilin Ou

Blockchain Infrastructure / Security / Agent Audit Layer



Shilin Ou

Secure infrastructure and
auditable agent systems

Relevant expertise

- SolarChain and blockchain infrastructure
- Secure data flows and audit design
- Agent reliability and technical-risk analysis
- System integration and prototype testing

Contribution highlight

Shilin brings SolarChain experience to a secure, modular, auditable evidence agent designed to extend beyond one prototype.

Expected contribution

- Co-develop the agent architecture with Sean.
- Design audit checkpoints for API keys, logs, citations, and outputs.
- Develop the secure-agent and technical-risk narrative.
- Test the MVP workflow and document failure modes.



DKU Team Profile

Governance / Open-Source Community / Developer Ecosystem



**Yutian (Claire)
Wang**

Governance and
developer-community
strategy

Related expertise

- DAO and decentralized governance research direction
- Open-source collaboration workflows
- Developer-community positioning and documentation

Contribution highlight

Claire brings a governance and developer-community perspective, moving the internal prototype toward a reproducible, open-source, public-facing AI application.

Key contributions

- Draft the open-source contribution and governance workflow.
- Review the GitHub structure, README, and public setup guide.
- Compare DevPost and hackathon examples for demo expectations.



DKU Team Profile

Interdisciplinary Framing / Open Science / Ecosystem Strategy



**Polina
Postnikova**

Public policy, user needs,
and community impact

Related expertise

- Data-to-policy research direction
- Public-policy interpretation and user needs
- Cross-border mobility and community impact

Contribution highlight

Polina brings a policy and user-community lens, translating historical evidence into public questions about Silk Road, Belt and Road, governance, and local communities.

Key contributions

- Draft user personas and policy-oriented demo questions.
- Review the UX walkthrough from a non-developer perspective.
- Prepare the SDGs, local-community, and public-value narrative.
- Separate evidence, interpretation, and modern analogy.





DKU Team Profile

Interdisciplinary Framing / Open Science / Ecosystem Strategy



**Prof. Luyao
Zhang**

Faculty Mentor

Contribution highlight

Mentored StableEval Atlas from incubation to external engagement, aligning the team's strengths with open science, responsible innovation, community impact in China, and technology frontiers.

Related expertise

- Responsible AI, blockchain, intelligent economics, and data governance
- Open science, SDGs, and global collaboration
- Standards, industry partnerships, Web3/AI, and open-source tools

Key contributions

- Shaped the vision, interdisciplinary framing, and intellectual coherence.
- Connected industry, developer, open-source, standards, and responsible-AI networks.
- Guided advising, proposal, deck, submission, seminars, and roadmap.



External Advisors

Acknowledgments · In alphabetical order by last name



Prof. Shiping Chen

Advising focus

- External academic advising
- Distributed-systems perspective
- Trustworthy infrastructure guidance
- Blockchain / Web3 architecture insight
- Research scalability feedback



Dr. Dongping Liu

Advising focus

- Industry advising
- AWS Open Data guidance
- Secure-agent architecture
- AI guardrails review
- MCP / cloud infrastructure input



Prof. Xuechao Wang

Advising focus

- External academic advising
- AI security perspective
- Agent safety and guardrails feedback
- Evaluation methodology guidance
- Responsible AI research input

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